

Physics - Question 81

Original paper page 17; original question and option regions

81. A parallel plate capacitor has 91 plates, all are identical and arranged with same spacing between them. If the capacitance between adjacent plates is 3pF . What will be the resultant capacitance ?

1) 273 pF

2) 30 pF

3) 94 pF

4) 270 pF

Ans : 4

Physics - Question 82

Original paper page 17; original question and option regions

82. A 1Ω resistance in series with an ammeter is balanced by 75 cm of potentiometer wire. A standard cell of emf 1.02 V is balanced by 50 cm. The ammeter shows a reading of 1.5A. Then the error in ammeter reading is

1) 0.03 A

2) 3 A

3) 1.3 A

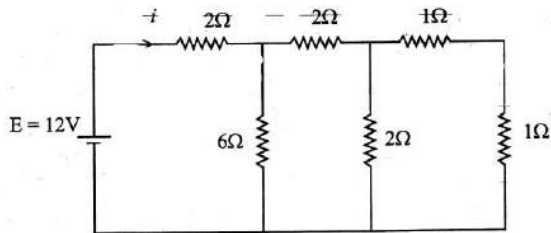
4) 0.3 A

Ans : 1

Physics - Question 83

Original paper page 17; original question and option regions

83. A battery of 12 V is connected in the circuit as shown in figure. Find the magnitude of current 'i'.



1) 1A

2) 2A

3) 3A

4) 4 A

Ans : 3

Physics - Question 84

Original paper page 18; original question and option regions

84. A rectangular copper coil is placed in uniform magnetic field of induction 40 mT with its plane perpendicular to the field. The area of the coil is shrinking at a constant rate of $0.5 \text{ m}^2\text{s}^{-1}$. The emf induced in the coil is

1) 10 mV

2) 20 mV

3) 80 mV

4) 40 mV

Ans : 2

Physics - Question 85

Original paper page 18; original question and option regions

85. The range of voltmeter is 10V and its internal resistance is 50Ω . To convert it to a voltmeter of range 15V , how much resistance is to be added ?

1) Add $25\ \Omega$ resistor in parallel

2) Add $25\ \Omega$ resistor in series

3) Add $125\ \Omega$ resistor in parallel

4) Add $125\ \Omega$ resistor in series

Ans : 2

Physics - Question 86

Original paper page 18; original question and option regions

86. A bar magnet has a coercivity of $4 \times 10^3 \text{ Am}^{-1}$. It is placed inside a solenoid of length 12cm having 60 turns in order to demagnetize it. The amount of current that should be passed through the solenoid is
- 1) 16 A 2) 8 A 3) 4 A 4) 2 A

Ans : 2

Physics - Question 87

Original paper page 18; original question and option regions

87. Match the following

Quantity

Formula

1) Magnetic flux linked with a coil

$$a) -N \frac{d\phi}{dt}$$

2) Induced emf

$$b) \mu_r \mu_0 n_1 n_2 \pi r_1^2 l$$

3) Force on a charged particle moving
In a electric and magnetic field

$$c) BA \cos \theta$$

4) Mutual inductance of a solenoid

$$d) q(\vec{E} + \vec{v} \times \vec{B})$$

1) 1-c, 2-d, 3-b, 4-a

2) 1-c, 2-a, 3-d, 4-b

3) 1-b, 2-a, 3-c, 4-d

4) 1-a, 2-b, 3-d, 4-c

Ans : 2

Physics - Question 88

Original paper page 18; original question and option regions

88. The electric current in A.C. circuit is given by the relation $i = 3 \sin \omega t + 4 \cos \omega t$. The rms value of the current in the circuit is in amperes

1) $\frac{5}{\sqrt{2}}$

2) $5\sqrt{2}$

3) $5\sqrt{2}$

4) $\frac{1}{\sqrt{2}}$

Ans : 1

Physics - Question 89

Original paper page 19; original question and option regions

89. 5% of the power of 200w bulb is converted into visible radiation. The average intensity of visible radiation at a distance of 1m from the bulb is

1) 0.5 w/m^2

2) 0.8 w/m^2

3) 0.4 w/m^2

4) 2 w/m^2

Ans : 2

Physics - Question 90

Original paper page 19; original question and option regions

90. When a light beam of frequency ν is incident on a metal, photo electrons are emitted. If these electrons describe a circle of radius r in a magnetic field of strength B , then the work function of the metal is

- 1) $h\nu + \frac{Ber}{2me}$ 2) $h\nu - \frac{(Ber)^2}{2me}$ 3) $h\nu + \frac{(Ber)^2}{2me}$ 4) $h\nu - \frac{Ber}{2me}$

Ans : 2 / no answer

Physics - Question 91

Original paper page 19; original question and option regions

91. The distance of closet approach of an α - particle fired at nucleus with momentum P is d. The distance of closest approach when the α - particle is fired at same nucleus with momentum 3P will be

1) 3d

2) d/3

3) 9 d

4) d/9

Ans : 4

Physics - Question 92

Original paper page 19; original question and option regions

92. In a nuclear reactor the number of U^{235} nuclei undergoing fissions per second is 4×10^{20} . If the energy released per fission is 250 MeV, the total energy released in 10 hours is $(1 \text{ eV} = 1.6 \times 10^{-19} \text{ J})$

1) $576 \times 10^6 \text{ J}$

2) $576 \times 10^{12} \text{ J}$

3) $576 \times 10^{15} \text{ J}$

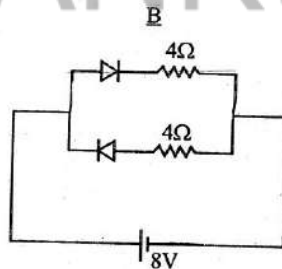
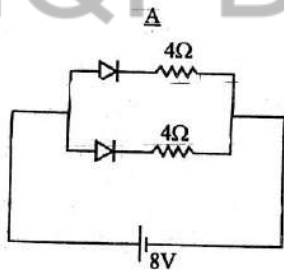
4) $576 \times 10^{18} \text{ J}$

Ans : 2

Physics - Question 93

Original paper page 19; original question and option regions

93. Current flowing in each of the following circuits A and B respectively are the forward resistance of diodes)



1) 1 A and 2 A

2) 2 A and 1A

3) 4 A and 2 A

4) 2 A and 4A

Ans : 3

Physics - Question 94

Original paper page 19; original question and option regions

94. A message signal of frequency 20 kHz is used to modulate a carrier of frequency 1.5 MHz. The side bands produced are

1) 520 Hz, 480 kHz

2) 1520 kHz, 1480 kHz

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Physics - Question 94 (continued)

Original paper page 20; original question and option regions



Physics - Question 95

Original paper page 20; original question and option regions

95. If energy (E), force (F) and linear momentum (P) and fundamental quantities, then match the following and give correct answer.

A

Physical quantity

a) Mass

b) Length

c) Time

1) a-d, b-e, c-f

B

Dimensional formula

d) $E^0 F^{-1} P^1$

e) $E^{-1} F^0 P^2$

$E^{-1} F^0 P^2$

3) a-e, b-f, c-d

4) a-e, b-d, c-f

Ans : 3

Physics - Question 96

Original paper page 20; original question and option regions

96. A vehicle moving with a constant acceleration from A to B in a straight line AB, has velocities 'u' and 'v' at A and B respectively. C is the mid point of AB. If time taken to travel from A to C is twice the time to travel from C to B then the velocity of the vehicle 'v' at B is
- 1) $5u$ 2) $6u$ 3) $7u$ 4) $8u$

Ans : 3

Physics - Question 97

Original paper page 20; original question and option regions

97. An object is thrown towards the tower which is at a horizontal distance of 50 m with initial velocity of 10ms^{-1} and making an angle 30° with the horizontal. The object the tower at certain height. The height from and bottom of the tower where the object the tower is ($g=10\text{ms}^{-2}$)

1) $\frac{50}{\sqrt{3}}\left[1-\frac{10}{\sqrt{3}}\right]m$ 2) $\frac{50}{3}\left[1-\frac{10}{\sqrt{3}}\right]m$ 3) $\frac{100}{\sqrt{3}}\left[1-\frac{10}{\sqrt{3}}\right]m$ 4) $\frac{100}{3}\left[1-\frac{10}{\sqrt{3}}\right]m$

Ans : 1

Physics - Question 98

Original paper page 20; original question and option regions

98. A body of mass 5 kg is projected with a velocity 'u' at an angle 45° to the horizontal. The magnitude of angular momentum of the body when it is at the highest position with respect to the point of projection is

1) $\frac{u^3}{5\sqrt{5}g}$

2) $\frac{5u^3}{4\sqrt{2}g}$

3) $\frac{u^3}{5g}$

4) $\frac{u^3}{\sqrt{5}g}$

Ans : 2

Physics - Question 99

Original paper page 21; original question and option regions

99. A balloon of mass 10kg is raising up with an acceleration of 20ms^{-2} . If a mass of 4 kg removed from the balloon its acceleration becomes (take $g=10\text{ms}^{-2}$)

1) 40ms^{-2}

2) 20ms^{-2}

3) 30ms^{-2}

4) 12ms^{-2}

Ans : 1

Physics - Question 100

Original paper page 21; original question and option regions

100. A uniform chain of length ' ℓ ' is placed on a smooth horizontal table, such that half of its length hangsover one edge. It is released from rest, the velocity with which it leaves the table is

1) $\sqrt{\frac{3gl}{4}}$

2) $\sqrt{\frac{3gl}{2}}$

3) $\sqrt{\frac{2gl}{3}}$

4) $\sqrt{\frac{gl}{3}}$

Ans : 1

Physics - Question 101

Original paper page 21; original question and option regions

101. Two cylindrical vessels of equal cross sectional area A , contain liquid of density ρ , upto the heights h_1 and h_2 . After inter connection the levels become equal then work done by force of gravity in this process is

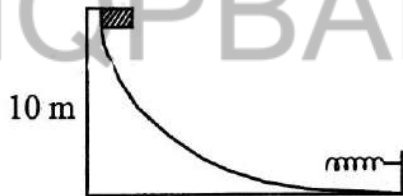
1) $\rho A \left(\frac{h_1 - h_2}{2} \right)^2 g$ 2) $\rho A \left(\frac{h_1 - h_2}{2} \right)$ 3) $\rho A \left(\frac{h_1 + h_2}{2} \right) g$ 4) $\rho A (h_1 - h_2) g$

Ans : 1

Physics - Question 102

Original paper page 21; original question and option regions

102. A smooth curved surface of height 10m is ended horizontal. A spring of force 200 Nm^{-1} is fixed at the horizontal end as shown in figure when an object of mass 10g is released from the top, it travels along the curved path and collides with the spring. The maximum compression in the spring is ($g = 10 \text{ ms}^{-2}$)



1) 10 m

2) 0.1 m

3) 1m

4) 0.01. m

Ans : 2

Physics - Question 103

Original paper page 21; original question and option regions

103. The ratio of the radii of gyration of the disc about its axis and about a tangent per to its plane will be

1) $\frac{1}{\sqrt{3}}$

2) $\sqrt{\frac{3}{2}}$

3) $\frac{1}{\sqrt{2}}$

4) $\sqrt{\frac{5}{3}}$

Ans : 1

Physics - Question 104

Original paper page 22; original question and option regions

104. A solid sphere of radius 'r' is rolling on a horizontal surface. The ratio between the kinetic energy and total energy

1) $\frac{5}{7}$

2) $\frac{2}{7}$

3) $\frac{1}{2}$

4) $\frac{1}{7}$

Ans : 2

Physics - Question 105

Original paper page 22; original question and option regions

105. A copper sphere attached to the bottom of a vertical spring is oscillating with time period 10sec. If the copper sphere is immersed in a fluid (assume the viscosity of the fluid is negligible) of specific gravity $\frac{1}{4}$ of that of the copper, then time period of the oscillation is

1) 5 sec

2) 10 sec

3) 2.5 sec

4) 20 sec

Ans : 2

Physics - Question 106

Original paper page 22; original question and option regions

106. Two spheres of masses 16 kg and 4 kg are separated by a distance 30 m on a table. Then the distance from sphere of mass 16 kg at which the net gravitational force becomes zero is

1) 10 m

2) 20 m

3) 15 m

4) 5m

Ans : 2

Physics - Question 107

Original paper page 22; original question and option regions

107. A soap bubble of diameter a is produced using the soap solution of surface tension ' T '. Find the energy required to double the radius of the bubble without change of temperature

1) $2\pi a^2 T$

2) $6\pi a^2 T$

3) $3\pi a^2 T$

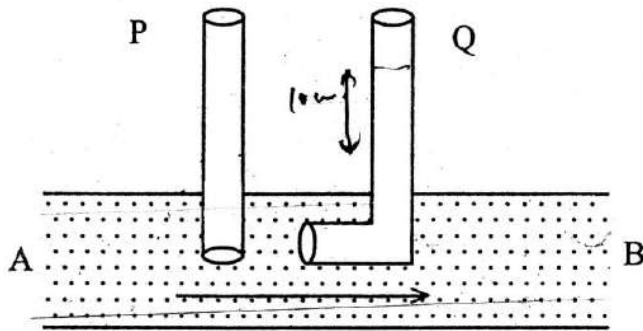
4) $12\pi a^2 T$

Ans : 2

Physics - Question 108

Original paper page 22; original question and option regions

108. A liquid flows along a horizontal pipe AB of uniform cross-section. The difference between the level of the liquid in tubes P and Q is 10cm. The diameter of the tubes P and Q the same. Then $(g = 9.8 \text{ ms}^{-2})$



- 1) Level in P is greater than that of Q and velocity of flow is 1.4 m/s

Physics - Question 108 (continued)

Original paper page 23; original question and option regions

4) Level in Q is greater than that of P and velocity of flow is 0.7 m/s

Ans : 2

Physics - Question 109

Original paper page 23; original question and option regions

109. A pan ;filled with hot food cool from 94°C to 86°C in 2 minutes. When the room temperature is at 20°C , the time taken to cool from 71°C to 69°C is

1) 50 sec

2) 40 sec

3) 38 sec

4) 42 sec

Ans : 4

Physics - Question 110

Original paper page 23; original question and option regions

110. Two metals rods of lengths, L_1 and L_2 and co-efficients of linear expansion α_1 and α_2 respectively are welded together to make a composite rod of length $(L_1 + L_2)$ at 0°C . Find the effective co-efficient of linear expansion of the composite rod.

1) $\frac{L_1\alpha_1^2 - L_2\alpha_2^2}{L_1^2 + L_2^2}$

2) $\frac{L_1^2\alpha_1 - L_2^2\alpha_2}{L_1^2 + L_2^2}$

3) $\frac{L_1\alpha_1 + L_2\alpha_2}{L_1 - L_2}$

4) $\frac{L_1\alpha_1 + L_2\alpha_2}{L_1 + L_2}$

Ans : 4

Physics - Question 111

Original paper page 23; original question and option regions

111. An ideal diatomic gas is heated at constant pressure. What fraction of heat energy is utilized to increase its internal energy ?

1) $\frac{5}{7}$

2) $\frac{2}{5}$

3) $\frac{3}{5}$

4) $\frac{3}{7}$

Ans : 1

Physics - Question 112

Original paper page 23; original question and option regions

112. If the pressure of the gas contained in a vessel is increased by 0.4%, when heated through 1°C . What is the initial temperature of the gas ?

1) 250K

2) 250°C

3) 2500K

4) 25°C

Ans : 1

Physics - Question 113

Original paper page 23; original question and option regions

113. Specific heat of one mole of Hydrogen at constant pressure and at constant volume are 450 JK^{-1} and 300 JK^{-1} respectively. Then what is the density of the gas at S.T.P

1) $(P_{\text{atm}} = 1.013 \times 10^5 \text{ Nm}^{-2})$

1) 2.47 kg/m^3

2) 3.0 kg/m^3

3) 3.58 kg/m^3

4) 4.0 kg/m^3

Ans : No answer

Physics - Question 114

Original paper page 24; original question and option regions

114. In a medium in which a transverse progressive wave travelling the phase different between the points with a separation of 1.25 cm is $\frac{\pi}{4}$. If the frequency of wave 1000 Hz, the velocity in the medium is

1) 10^4 ms^{-1}

2) 125 ms^{-1}

3) 100 ms^{-1}

4) 10 ms^{-1}

Ans : 3

Physics - Question 115

Original paper page 24; original question and option regions

115. Two wires of the same diameter and same material have length 1.2 m and 2m, vibration with the same fundamental frequency. If the shorter wire is stretched by a force 36 kg wt, then the tension in the longer wire is

1) 50 kg wt

2) 200 kg wt

3) 100kg wt

4) 400 kg wt

Ans : 3

Physics - Question 116

Original paper page 24; original question and option regions

116. The object and the image are at distance of 9 cm and 16 cm respectively from the focus of an equiconvex lens of radius of curvature 12 cm. The refractive index of the material of the lens is

1) 1.45

2) 1.5

3) 1.55

4) 1.6

Ans : 2

Physics - Question 117

Original paper page 24; original question and option regions

117. A plane wave of wavelength $6250 \text{ \AA}$ is incident normally on a slit of width $2 \times 10^{-2} \text{ cm}$. The width of the principle maximum of diffraction pattern on a screen at a distance of 50 cm will be

- 1) $312 \times 10^{-3} \text{ cm}$ 2) $312.5 \times 10^{-4} \text{ cm}$ 3) 312 cm 4) $312.5 \times 10^{-5} \text{ cm}$

Ans : 1

Physics - Question 118

Original paper page 24; original question and option regions

118. A Gaussian surface in the cylinder of cross section ' πa^2 ' and length 'L' is immersed in a uniform electric field $\vec{E}$ with the cylinder axis parallel to the field. The flux ' ϕ ' of the electric field through the closed surface is

1) $2\pi a^2 \vec{E}$

2) $\pi a^2 \vec{E}L$

3) $\pi a^2 (2 + L) \vec{E}$

4) zero

Ans : 4

Physics - Question 119

Original paper page 24; original question and option regions

119. A total charge of $5\mu\text{C}$ is distributed uniformly on the surface of the thin walled semispherical cup. If the electric field strength at the centre of the semisphere is $9 \times 10^8 \text{ NC}^{-1}$ the radius of the cup is

$$\left(\frac{1}{4\pi \epsilon_0} = 9 \times 10^9 \text{ Nm}^2\text{C}^{-2} \right)$$

1) 5 mm

2) 10 mm

3) 5 cm

4) 10 cm

Physics - Question 120

Original paper page 25; original question and option regions

120. The potential difference between the plates of a capacitor is increased by 20%. The energy stored on the capacitor increases by

1) 20%

2) 22%

3) 40%

4) 44%

Ans : 4

CHEMISTRY